

Dissertation Title:
**A Privacy-Preserving Federated Learning Framework
with Explainability Auditing for Staffing-Acuity
Mismatch Prediction in Long-Term Care**

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1. Broad Area of Work

The broad area of this dissertation lies at the intersection of Privacy-Preserving Machine Learning, Healthcare Artificial Intelligence, Federated Learning (FL), Explainable AI (XAI), and Long-Term Care (LTC) Operations Research.

The specific application domain is the senior living and long-term care industry, a sector that encompasses Memory Care, Skilled Nursing Facilities (SNF), and Independent Living communities. Despite accumulating years of rich operational and resident-acuity data, these facilities operate in informational silos due to HIPAA-mandated restrictions on Protected Health Information (PHI) sharing. This structural isolation makes centralised machine learning infeasible across facilities.

FedAcuity addresses this gap by applying Federated Learning - a distributed ML paradigm in which model weights, rather than raw data, are shared - to the prediction of staffing-acuity mismatch in LTC settings. The dissertation further contributes a synthetic data benchmark validated against the MIMIC-IV clinical dataset, and an original 4-dimension XAI Audit Scorecard to evaluate the trustworthiness of federated models in a clinical care context.

The research cuts across the following academic and applied areas:

- Federated Learning & Distributed ML Algorithms (FedAvg, FedProx, Clustered FL)
- Differential Privacy in Machine Learning (Opacus / DP-SGD)
- Synthetic Clinical Data Generation & Validation (CTGAN, MIMIC-IV, SDV)
- Explainable Artificial Intelligence in Healthcare (SHAP, LIME, XAI audit frameworks)
- Long-Term Care Operations & Workforce Planning (MDS/RUG acuity scoring, ADL dependency metrics)
- Healthcare Compliance & Privacy Law (HIPAA, PHI governance in ML applications)

2. Background

2.1 A Systemic Staffing Crisis in Long-Term Care

Long-term care (LTC) facilities - including skilled nursing facilities, memory care units, and independent living communities - are experiencing a documented and deepening staffing crisis. According to the American Health Care Association (AHCA, 2022), 87% of nursing homes report moderate-to-high staffing shortages. Residents in these settings average approximately 8.5 medications per person, and their care needs, quantified through Activities of Daily Living (ADL) dependency scores, cognitive status, fall risk, and clinical acuity shift dynamically on a daily basis.

Staffing-acuity mismatch occurs when the nursing hours available at a facility fall short of the care demand generated by the current resident population. This mismatch is the root cause of adverse outcomes: medication errors, preventable falls, delayed clinical interventions, and regulatory violations. Despite years of operational data being logged in Electronic Health Records (EHRs) and acuity management systems, no cross-facility predictive tool exists specifically for staffing-acuity mismatch prediction in LTC.

2.2 The HIPAA Barrier and Why Centralised ML Is Not Feasible

Resident records in LTC facilities constitute Protected Health Information (PHI) under the Health Insurance Portability and Accountability Act (HIPAA). Cross-facility data sharing for model training purposes is not legally permissible without resident consent - a consent mechanism that does not exist at scale for retrospective ML training. HIPAA penalties can reach \$1.9 million per violation category. This legal constraint makes centralised multi-facility machine learning impossible under current regulation.

2.3 Federated Learning as the Solution

Federated Learning (FL), introduced by McMahan et al. (2017) as the FedAvg algorithm, offers a principled solution to this dilemma. In FL, each participating facility trains a local model on its own data. Only model weights - mathematical parameters containing no patient identifiers - are transmitted to a central aggregation server. The server merges these weights and distributes an improved global model back to all clients. Raw data

never leaves the facility.

While FL has been applied extensively in hospital settings for tasks such as readmission prediction, sepsis detection, and diagnostic imaging, its application to LTC facilities - with their distinct longitudinal data structures, extreme non-IID heterogeneity across care types, and specific staffing-acuity prediction target has not been studied. FedAcuity fills this documented gap.

2.4 Relevance to the Work Environment

This dissertation is grounded in the researcher's professional experience in the Senior Living and Long-Term Care domain, with direct exposure to acuity scoring systems (MDS/RUG), staffing operations, EHR data schemas, and the operational challenges of workforce planning at scale. The research is not theoretical: the synthetic dataset is schema-designed based on observed real-world LTC data structures (without any PHI), the prediction target operationalises a real staffing decision problem, and the XAI audit dimensions are designed for clinical plausibility by healthcare workforce standards.

3. Objectives

The primary research question guiding this dissertation is:

"Can federated learning enable accurate and privacy-preserving prediction of staffing-acuity mismatch in long-term care settings, while providing clinically trustworthy and fair model explanations?"

The dissertation is structured around three standalone, independently publishable contributions, each addressing a specific research sub-objective:

Objective 1: Engineering the FedAcuity FL System

- Design and implement a domain-driven clustered federated learning (CFL) strategy tailored to the extreme non-IID data heterogeneity of LTC facilities (Memory Care vs. Skilled Nursing vs. Independent Living).
- Implement and comparatively evaluate five model variants: Local (no federation), Centralised (oracle), FedAvg (McMahan et al. 2017), FedProx (Li et al. 2020), and Clustered FL.

- Integrate a differential privacy (DP) layer using Opacus (DP-SGD) and analyse the privacy-utility tradeoff across epsilon values $\epsilon \in \{1, 2, 5, 10, \infty\}$.
- Demonstrate that clustered FL significantly outperforms standard FedAvg on AUC-ROC for the mismatch prediction task on non-IID LTC data.

Objective 2: Synthetic Data Benchmark Creation and Validation

- Generate a synthetic LTC dataset representing 10 simulated facilities (3 Memory Care, 4 Skilled Nursing, 3 Independent Living) with 3 years of longitudinal daily records per facility (~109,500 total rows) using CTGAN (Conditional Tabular GAN via SDV library).
- Engineer deliberate non-IID distributional differences across care types to reflect real operational heterogeneity in LTC.
- Validate the synthetic dataset's statistical fidelity against the MIMIC-IV clinical dataset (elderly subset: age ≥ 65 , 3+ comorbidities) using KS-test, Frobenius norm comparison, and Train-on-Synthetic Test-on-Real (TSTR) experiments.
- Publish the validated benchmark dataset on GitHub and Zenodo as an open, reusable research resource - the first public FL-ready benchmark for the senior living domain.

Objective 3: XAI Audit Scorecard for Federated LTC Models

- Design and implement a 4-dimension XAI Audit Scorecard for federated models, comprising: D1 - Explanation Fidelity (SHAP rank correlation vs. centralised oracle), D2 - Explanation Stability (perturbation-based SHAP shift testing), D3 - Subgroup Fairness (equalized odds and demographic parity across care types), D4 - Clinical Plausibility (expert literature comparison of top SHAP features).
- Apply the scorecard to all five model variants and demonstrate that the clustered FL approach produces superior XAI audit scores compared to standard FedAvg.
- Produce a normalised, radar-chart-visualised XAI Audit Scorecard as a reusable framework for clinical AI explainability assessment in federated healthcare settings.

4. Scope of Work

The scope of this dissertation encompasses the following work to be independently executed by the student:

- Full design, implementation, and evaluation of the FedAcuity federated learning simulation using the Flower (flwr) framework with 10 simulated facility clients.
- Synthetic LTC dataset generation using CTGAN (SDV library), encompassing schema design, non-IID engineering, label definition, and data validation against MIMIC-IV.
- Implementation of all five model variants (Local, Centralised, FedAvg, FedProx, Clustered FL) with full comparative evaluation.
- Differential privacy integration via Opacus with epsilon-sweep analysis and privacy-utility tradeoff reporting.
- 4-dimension XAI audit pipeline: SHAP computation, fidelity testing, stability perturbation testing, fairness metrics, and clinical plausibility review.
- Statistical validation: Mann-Whitney U tests, bootstrap confidence intervals, and Fisher exact tests on key results.
- All visualisations including convergence curves, SHAP beeswarm plots, privacy-utility tradeoff curves, and XAI radar chart.
- Full dissertation write-up across all chapters (Introduction, Literature Review, Methodology, Results, Discussion, Conclusion).
- Research paper draft targeting a peer-reviewed venue (IEEE Journal of Biomedical and Health Informatics or equivalent).

Out of scope for this dissertation: Clinical trials or real patient data collection; direct integration with live EHR production systems; deployment to production LTC facility infrastructure; legal or regulatory compliance implementation.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	25 Apr 2026 - 8 May 2026	Literature review, system architecture design, environment setup.
Data Engineering	9 May 2026 - 21 May 2026	Synthetic dataset generation, validation, and dataset release.
FL Implementation & DP	22 May 2026 - 18 Jun 2026	FL simulation with 5 model variants, differential privacy sweep, model comparison.
XAI Audit Engine	19 Jun 2026 - 9 Jul 2026	SHAP pipeline, fidelity, stability, fairness and clinical plausibility audit.
Dissertation Review	10 Jul 2026 - 27 Jul 2026	Full dissertation draft, methodology, results, discussion and plagiarism check.
Submission	28 Jul 2026 - 2 Aug 2026	Final review and submission of dissertation along with supervisor evaluation report.

6. Literature References

The following literature forms the foundational base for FedAcuity, organised thematically across the three contribution areas.

6.1 Federated Learning - Core Algorithms

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6.2 Federated Learning in Healthcare

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6.3 Long-Term Care Domain & Staffing Research

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6.4 Synthetic Data Generation & Clinical Data Validation

- [14] Jordon, J., Szpruch, L., Houssiau, F., et al. (2022). Synthetic Data - what, why and how? The Alan Turing Institute. [Foundation for synthetic data methodology]
- [15] Xu, L., Skoularidou, M., Cuesta-Infante, A., & Veeramachaneni, K. (2019). Modeling Tabular data using Conditional GAN. NeurIPS 2019. [CTGAN - core data generation method]
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- [17] Johnson, A.E.W., et al. (2023). MIMIC-IV, a freely accessible electronic health record dataset. Scientific Data, 10, 1. [MIMIC-IV - fidelity validation anchor]

6.5 Explainable AI in Healthcare

- [18] Lundberg, S.M., & Lee, S.I. (2017). A Unified Approach to Interpreting Model Predictions (SHAP). NeurIPS 2017. [Core XAI method for C3]
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6.6 Differential Privacy

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7. Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
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8. Remarks of the Supervisor

FedAcuity addresses a significant gap in federated learning for long-term care AI. The research question is clearly framed with three independently evaluable contributions and a realistic plan. The framework demonstrates strong technical rigour, including clustered federation for non-IID data heterogeneity, CTGAN + MIMIC-IV synthetic validation, and a structured XAI audit scorecard - with clear potential for peer-reviewed publication.

Grounded in the student's professional LTC experience, I approve this project as suitable for the M.Tech AI/ML dissertation at BITS Pilani WILP.

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2025-2026

AIMLCZG628T : DISSERTATION OUTLINE

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